



Project: SQL Exploratory Data Analysis

A structured deep-dive into our new data warehouse — uncovering structure, data quality, and actionable business insights across Sales, Customers, and Products.

SUMMARY REPORT

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Problem Statement	Project Objective & Scope	Database Exploration
Why this analysis was needed	What we set out to accomplish	Structure, schema, tables relationship, and data quality
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Customer, Product, Geography, and Time	Key Metrics (KPI) determination	Performance, trends, and strategic insights

The Challenge: A Data Warehouse Without Visibility

Following the migration to **AdventureWorksDW2025**, management lacked the understanding of its structure, data quality, and business value.



1

No Sales Visibility

Revenue trends, top performers, and regional breakdowns were unknown or undefined.

2

Unknown Customer Behavior

Customer demographics, geography, and purchasing patterns had not yet been identified or explored.

3

Unclear Product Performance

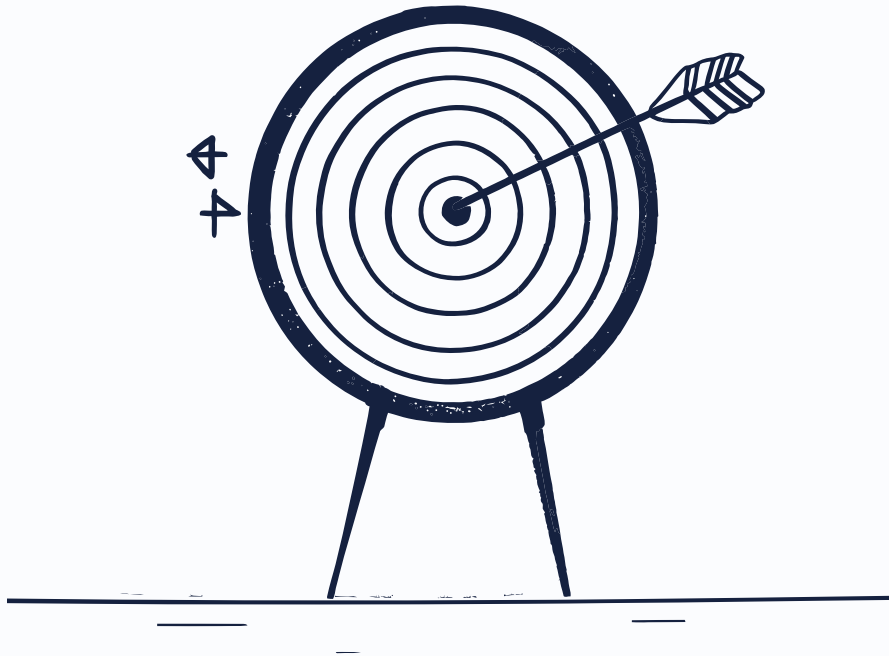
No established hierarchy or ranking existed for categories, subcategories, or individual SKUs.

4

Unclear Data Relationships

Fact and dimension table linkages were unknown, making cross-entity analysis unreliable.

What we set out to accomplish?



Primary Objective

Explore, understand, and extract meaningful business insights from the AdventureWorksDW2025 data warehouse using structured SQL analysis.

Scope of Analysis

- Understand database structure and relationships
- Analyze customer demographics and behavior
- Evaluate product and category performance in terms of sales
- Measure key business KPIs
- Identify sales trends over time (YoY, MoM)
- Analyze geographical performance in terms of sales or revenue

Understanding the Data Warehouse Structure

Before extracting insights, we systematically mapped the architecture of AdventureWorksDW2025 to ensure data reliability and relational integrity.



Tables & Views

All tables and views were identified. In Total, 31 Table and 5 Views were found. However, not all tables were relevant.



Columns Profiling

Column names, data types, and nullability examined across all key tables to assess completeness and consistency.



Keys & Relationships

Primary and foreign keys identified per table, establishing a reliable relational map for cross-entity joins.



Fact vs. Dimension

Tables classified using naming convention standards as well as by through checking of an individual table. Two fact tables related to sales were found: FactInternetSales & FactResellerSales

Profiling the Four Core Dimensions



Customer

Demographics, geography, and segmentation across all buyer profiles were revealed.



Product

Full hierarchy mapped:
Category → Subcategory → Product. Attributes including cost and specifications reviewed.



Geography

Sales and customer distribution mapped across countries and cities to identify regional concentration.



Time

Order date range, shipping time, and delivery gaps analyzed. Inserted dates were logically correct.

Core Business Measures at a Glance

SQL queries extracted the following headline KPIs from the data warehouse, providing the foundation for all performance analysis.

18.5K

Total Customers

Unique customer records profiled across all regions

606

Total Products

SKUs spanning multiple categories and subcategories

\$109M

Total Sales

Aggregate revenue recorded across all order periods

\$802

Avg. Product Cost

Mean cost across all products

Product Performance Analysis in terms of Sales

Revenue was assessed at the Category, Subcategory, and Product levels, enabling precise identification of top and bottom performers.

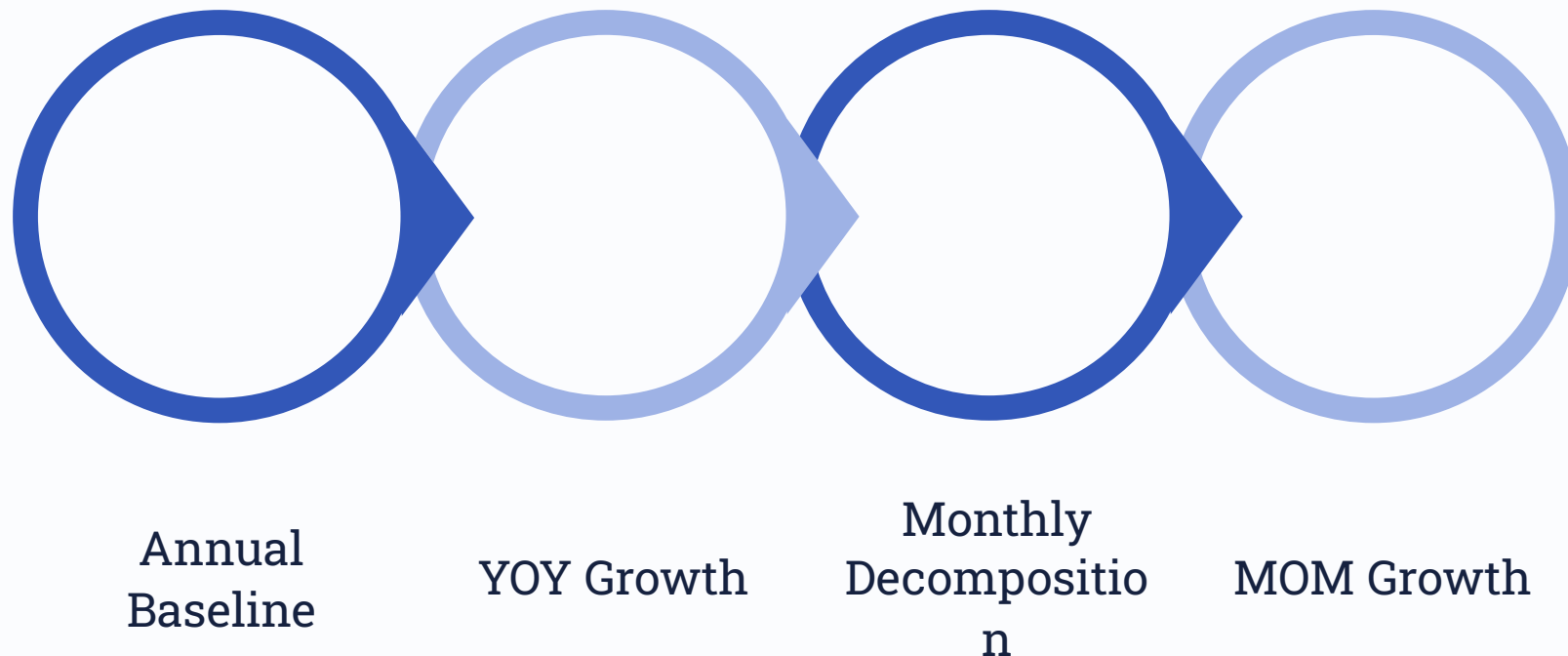
✅ **Top Performer:** Bikes dominated revenue, contributing the largest share across all product categories. Top 5 Products are Mountain-200 Black 38, Mountain-200 Black 42, Mountain-200 Silver 38, Mountain-200 Silver 42, Mountain-200 Silver 46

⚠️ **Bottom Performers:** Accessories and clothing SKUs showed minimal revenue contribution. Bottom 5 SKUs are LL Road Seat/Saddle, Mountain Bike Socks L, LL Touring Frame - Blue 58, LL Mountain Frame - Black 40, LL Touring Seat/Saddle.



Year-over-Year & Month-over-Month Sales Growth

Temporal analysis revealed steady annual growth, but low seasonality, offering a foundation for forward-looking planning.



① **YOY Trend:** Sales demonstrated consistent year-over-year growth across the full date range, with acceleration in later periods. Total Sales increased by 34.69% in 2012 compared to 2011 and 46.68% in 2013 compared to 2012

① **MoM Insight:** There is no strong signal that shows that our revenue depends on seasonality, rather our analysis finds that Month-over-Month Sales fluctuate a lot.

Key Takeaways & Recommended Actions

→ Data Warehouse is Structurally Sound

The star schema is validated. Fact and dimension tables are properly linked and ready for advanced analytics and BI tooling.

→ Sales or Revenue Growth

Sales demonstrated consistent year-over-year growth across the full date range, with acceleration in later periods.

→ Bikes Drive Revenue — Accessories Lag

Product portfolio decisions should prioritize high-performing categories while evaluating the strategic fit of bottom-5 SKUs.

→ Build on This Foundation

Connect AdventureWorksDW2025 to a BI dashboard (e.g., Power BI) to democratize access to these insights across all business units.

